

Fulbright University Vietnam
CS313 Deep Learning for Artificial Intelligence

Final Project Report

Time-Series Data and Application to Stock Markets

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Introduction

Stock market forecasting is one of the most challenging applications of machine learning, owing to the non-stationary, high-noise nature of financial time-series data. This project applies deep learning methods — primarily Long Short-Term Memory (LSTM) networks — to two distinct markets: the US Nasdaq exchange and the Vietnamese stock market (HOSE/HNX). The work spans four interconnected tasks: price forecasting (Tasks 1 and 2), trading-signal identification (Task 3), and portfolio construction with risk management (Task 4).

The project was completed individually as part of the CS313 Deep Learning for Artificial Intelligence course at Fulbright University Vietnam. All modeling decisions, data-handling choices, and evaluation strategies are described in this report so that the reasoning behind each design choice is clear and reproducible.

Objective

The primary objectives of this project are to:

1. Build and evaluate multi-feature, multi-horizon LSTM models for stock price prediction on both Nasdaq and Vietnamese datasets.
2. Develop classification models to identify buying and selling signals in the Vietnamese market.
3. Construct a portfolio management pipeline that combines profitability prediction with risk scoring to produce actionable asset-allocation recommendations.

Report Structure

Section 2 describes the datasets and preprocessing pipeline. Section 3 covers Nasdaq price forecasting. Section 4 covers Vietnamese price forecasting. Section 5 details trading-signal identification. Section 6 describes portfolio construction and risk management. Section 7 discusses findings and limitations, and Section 8 concludes the report.

Data and Preprocessing

Datasets

Nasdaq Dataset. The Nasdaq dataset contains historical OHLCV data (Open, High, Low, Close, Adjusted Close, and Volume) for 1,564 tickers. Three representative tickers — AAPL, MSFT, and NVDA — were selected for model development, as they provide more than ten years of continuous trading history and represent diverse sectors within the technology industry.

Vietnamese Dataset. The Vietnam dataset covers 1,629 tickers from HOSE and HNX, with historical OHLCV data and TradingDate fields. Six tickers were selected for Tasks 2–4: **VCB** (banking), **HPG** (steel/manufacturing), **FPT** (technology), **VNM** (consumer staples), **MSN** (consumer goods), and **MWG** (retail). All six have more than 2,000 trading days of history. Supplementary data — financial ratios (P/E, ROE, ROA), dividend history, and industry-analysis reports — were also available but used only as context for portfolio-tier decisions rather than direct model features.

Feature Engineering

Raw OHLCV data was enriched with the following technical indicators, computed on a rolling basis to avoid any look-ahead leakage:

- **Simple Moving Averages (SMA-5, SMA-10, SMA-20):** capture short-, medium-, and longer-term trend direction.
- **Exponential Moving Averages (EMA-12, EMA-26):** give more weight to recent prices; used in MACD computation.
- **MACD and MACD Signal:** momentum oscillator measuring the divergence between EMA-12 and EMA-26.
- **Relative Strength Index (RSI-14):** overbought/oversold momentum indicator bounded in $[0, 100]$.
- **Bollinger Bands (upper, lower):** volatility bands $\pm 2\sigma$ around a 20-day SMA.

The Nasdaq models use 13 features; the Vietnam models use 12 features (identical set, adapted for the local price column naming). All features were verified to be computable from information available up to day t before being used as inputs to windows ending at day t .

Normalisation and Windowing

A **per-window MinMax normalisation** strategy was adopted. For each sliding window of 30 trading days, every feature column is independently scaled to $[0, 1]$ using the minimum and maximum of that window:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min} + \epsilon}, \quad \epsilon = 10^{-8}.$$

This scheme avoids global data leakage (a global scaler fitted on the training set would still contaminate the validation and test sets through its statistics) and handles the non-stationary drift typical of stock prices. Labels (future Adjusted Close values) are

normalised using the same window's Adjusted Close min/max and denormalised before metric computation.

Data Splitting

All splits are **strictly chronological**. No shuffling is performed at any point. The split ratios are 70% training, 15% validation, and 15% test, applied by window count. Standard k -fold cross-validation is not used; instead, a rolling-window scheme is applied on the validation set for hyperparameter selection.

Task 1: Nasdaq Stock Price Prediction

Task 1.1 — Multi-Feature Extension

The baseline sample code predicted Adjusted Close prices using only a single feature (Open price). Task 1.1 extended this to all 13 engineered features. The model input shape became (30, 13) — a 30-day window across 13 features — and the LSTM architecture was kept minimal to isolate the effect of the multi-feature extension:

$$\text{Input}(30, 13) \rightarrow \text{LSTM}(64) \rightarrow \text{Dense}(1)$$

Training used the Adam optimiser (learning rate 10^{-3}), MSE loss, batch size 64, and 10 epochs with `ModelCheckpoint` saving the best `val_loss` epoch.

Results for AAPL (Task 1.1, $k = 1$):

Table 1: Task 1.1 evaluation metrics for AAPL on the test set.

Ticker	MSE (norm)	MSE (\$ ²)	RMSE (\$)	MAE (\$)
AAPL	0.02457	4.0220	2.0055	1.2329
MSFT	0.02900	13.7469	3.7077	2.4313
NVDA	0.02619	33.6673	5.8024	3.8298

Adding multiple features consistently improved the normalised MSE compared to the single-feature Conv1D baseline provided in the sample notebook (baseline: ≈ 0.033). The richer feature set provides the LSTM with trend, momentum, and volatility signals that the Open price alone cannot convey.

Task 1.2 — k^{th} -Day-Ahead Forecast

Task 1.2 extended the pipeline to forecast the price k days into the future rather than just the next day. The same architecture and feature set from Task 1.1 were retained; only

the `horizon` parameter changed. Models were trained for $k \in \{3, 7\}$ for each of the three tickers.

Key observation: Forecast error grows monotonically with k for all three tickers. For AAPL, RMSE rises from \$2.01 at $k = 1$ to \$4.92 at $k = 7$ — a $2.45\times$ increase. MSFT and NVDA show even larger absolute growth, consistent with their higher absolute price levels and greater intraday volatility.

Table 2: Task 1.2 multi-horizon results for AAPL, MSFT, and NVDA.

Ticker	k	MSE (norm)	MSE (\$ ²)	RMSE (\$)	MAE (\$)
AAPL	1	0.02457	4.0220	2.0055	1.2329
	3	0.07641	11.2892	3.3599	2.1114
	7	0.18307	24.1643	4.9157	3.2361
MSFT	1	0.02900	13.7469	3.7077	2.4313
	3	0.07581	34.9734	5.9138	3.9359
	7	0.17408	85.4983	9.2465	6.2349
NVDA	1	0.02619	33.6673	5.8024	3.8298
	3	0.07203	90.0533	9.4896	6.3835
	7	0.17489	210.9241	14.5232	9.7407

Task 1.3 — k -Day Consecutive Forecast (Sequence Output)

Task 1.3 extended the model to predict all k future days simultaneously, rather than a single future day. The label became a length- k vector $[\text{Close}_{t+1}, \dots, \text{Close}_{t+k}]$. The model head changed from `Dense(1)` to `Dense( $k$ )`, while the LSTM trunk, normalisation, and training recipe remained unchanged.

This approach differs fundamentally from Task 1.2: Task 1.2 trains k independent single-step models, while Task 1.3 trains one multi-output model that jointly forecasts all steps. The per-step RMSE analysis (Table 4) reveals the error growth curve directly: step-1 RMSE closely matches the Task 1.2 $k = 1$ baseline (e.g. AAPL step-1 RMSE = \$2.014 vs. Task 1.2 $k = 1$ RMSE = \$2.006), and step-7 RMSE approaches the corresponding single-step $k = 7$ forecast. For NVDA at $k = 3$, the per-step RMSE values are \$5.844 / \$7.761 / \$9.352 (MAE: \$3.842 / \$5.173 / \$6.323), confirming the same monotone growth pattern across all tickers.

Table 3 shows the aggregate metrics and Table 4 shows per-step RMSE / MAE for all tickers at $k = 7$ (the harder horizon).

Table 3: Task 1.3 aggregate metrics for all tickers $\times k \in \{3, 7\}$.

Ticker	k	MSE (norm)	RMSE (\$)	MAE (\$)
AAPL	3	0.05159	2.7421	1.7039
	7	0.10261	3.7406	2.3318
MSFT	3	0.05339	4.9395	3.2237
	7	0.09899	6.8791	4.4785
NVDA	3	0.04848	7.7857	5.1129
	7	0.09728	10.8372	6.9921

Table 4: Task 1.3 per-step RMSE (\$) / MAE (\$) for all tickers at $k = 7$.

Ticker	Metric	S1	S2	S3	S4	S5	S6	S7
AAPL	RMSE	2.014	2.753	3.262	3.767	4.111	4.539	4.896
	MAE	1.256	1.741	2.077	2.421	2.653	2.961	3.213
MSFT	RMSE	3.766	4.908	6.016	6.776	7.529	8.654	8.904
	MAE	2.518	3.259	3.990	4.596	5.141	5.801	6.044
NVDA	RMSE	5.724	7.739	9.429	10.917	11.889	12.958	14.564
	MAE	3.742	5.243	6.298	7.233	7.932	8.766	9.730

Task 2: Vietnam Stock Price Prediction

Task 2.1 — Multi-Feature Extension

The same architecture and feature-engineering pipeline used for Nasdaq were applied to the six Vietnamese tickers (VCB, HPG, FPT, VNM, MSN, MWG), replacing the Adjusted Close column with Close (the standard label in the Vietnam dataset). The 12-feature input vector includes the raw OHLCV columns and all technical indicators described in Section 2.

Table 5: Task 2.1 evaluation metrics for the six Vietnam tickers ($k = 1$).

Ticker	MSE (norm)	MSE	RMSE	MAE
VCB	0.02944	2,420,503	1,555.80	1,196.34
HPG	0.02234	556,598	746.05	567.20
FPT	0.02857	1,958,136	1,399.33	930.86
VNM	0.03471	1,740,507	1,319.28	941.78
MSN	0.02990	8,152,611	2,855.28	2,093.54
MWG	0.02988	2,919,076	1,708.53	1,225.09

Task 2.2 — k^{th} -Day-Ahead Forecast

Mirroring Task 1.2, the Vietnam pipeline was run at $k \in \{1, 3, 7\}$ for all six tickers. The cross-market comparison reveals that Vietnam stocks exhibit higher relative RMSE growth with k than Nasdaq stocks, likely reflecting the higher volatility of the emerging Vietnamese market and sparser liquidity for some tickers.

Table 6: Task 2.2 multi-horizon results. All RMSE values in local currency units.

Ticker	k	MSE (norm)	MSE	RMSE	MAE
VCB	1	0.02944	2,420,503	1,555.80	1,196.34
	3	0.08926	6,690,905	2,586.68	1,966.72
	7	0.19170	15,345,048	3,917.28	3,072.91
HPG	1	0.02234	556,598	746.05	567.20
	3	0.06844	1,660,294	1,288.52	993.07
	7	0.20304	4,018,083	2,004.52	1,548.58
FPT	1	0.02857	1,958,136	1,399.33	930.86
	3	0.09221	5,536,871	2,353.06	1,627.68
	7	0.21846	12,032,503	3,468.79	2,476.41
VNM	1	0.03471	1,740,507	1,319.28	941.78
	3	0.10327	4,658,866	2,158.44	1,581.15
	7	0.21521	10,412,706	3,226.87	2,434.31
MSN	1	0.02990	8,152,611	2,855.28	2,093.54
	3	0.09896	26,233,418	5,121.86	3,833.58
	7	0.24446	56,196,520	7,496.43	5,805.26
MWG	1	0.02988	2,919,076	1,708.53	1,225.09
	3	0.10465	9,574,669	3,094.30	2,372.89
	7	0.36003	28,796,156	5,366.20	4,231.24

Task 2.3 — k -Day Consecutive Forecast

The multi-step output approach from Task 1.3 was replicated for Vietnam. The `Dense( $k$ )` head was applied with $k \in \{3, 7\}$ for all six tickers. Results are stored in `results_vn_kday[(ticker, k)]` and reused directly by Task 4 (the portfolio layer consumes the 5th step of the $k = 7$ model as the 5-day-forward expected return for each ticker).

Table 7: Task 2.3 aggregate metrics for all Vietnam tickers $\times k \in \{3, 7\}$.

Ticker	k	MSE (norm)	RMSE (VND)	MAE (VND)
VCB	3	0.05736	2,081.26	1,527.11
	7	0.11040	2,864.02	2,125.11
HPG	3	0.04618	1,058.60	788.26
	7	0.10813	1,497.82	1,108.74
FPT	3	0.06487	2,001.87	1,363.32
	7	0.11747	2,661.35	1,807.46
VNM	3	0.07042	1,833.59	1,296.19
	7	0.13552	2,458.53	1,787.71
MSN	3	0.06187	4,010.58	2,920.99
	7	0.13069	5,542.39	4,070.52
MWG	3	0.07315	2,629.12	1,918.90
	7	0.17478	3,891.90	2,864.59

Table 8: Task 2.3 per-step RMSE (VND) at $k = 7$ for all Vietnam tickers.

Ticker	Metric	S1	S2	S3	S4	S5	S6	S7
VCB	RMSE	1,573	2,129	2,562	2,904	3,159	3,394	3,730
	MAE	1,184	1,597	1,944	2,217	2,437	2,606	2,891
HPG	RMSE	814	1,077	1,298	1,479	1,667	1,836	1,964
	MAE	627	814	995	1,127	1,277	1,404	1,519
FPT	RMSE	1,412	2,014	2,353	2,656	2,949	3,210	3,455
	MAE	945	1,389	1,635	1,868	2,082	2,271	2,462
VNM	RMSE	1,410	1,902	2,225	2,448	2,704	2,943	3,128
	MAE	1,013	1,381	1,636	1,838	2,036	2,244	2,365
MSN	RMSE	2,879	4,099	5,016	5,580	6,108	6,686	7,186
	MAE	2,139	3,096	3,781	4,243	4,636	5,062	5,537
MWG	RMSE	2,092	2,653	3,232	3,909	4,056	5,007	5,231
	MAE	1,519	1,972	2,479	3,023	3,108	3,895	4,056

The step-by-step error growth follows the same monotone pattern observed in Task 1.3. Notably, MSN shows the largest absolute RMSE values (step-7: VND 7,186) owing to its higher price level and volatility, while HPG and FPT exhibit the most gradual growth curves relative to their step-1 baselines.

Task 3: Trading Signal Identification

Labelling Strategy

Buy and sell signals were formulated as a **future-return threshold** problem. A window ending at day t is labelled as a positive buying opportunity ($y = 1$) if the 5-day-forward return exceeds $+2\%$:

$$y_{\text{buy}} = \begin{cases} 1 & \text{if } \frac{\text{Close}_{t+5}}{\text{Close}_t} - 1 > +0.02 \\ 0 & \text{otherwise} \end{cases}$$

Similarly, a selling signal is positive if the 5-day-forward return falls below -2% :

$$y_{\text{sell}} = \begin{cases} 1 & \text{if } \frac{\text{Close}_{t+5}}{\text{Close}_t} - 1 < -0.02 \\ 0 & \text{otherwise} \end{cases}$$

The parameters $h = 5$ and $\tau = 0.02$ were chosen so that the positive class appears in 27–36% of windows (see Table 9), preventing class imbalance from trivially collapsing the classifier to “always predict 0.” Rule-based labels (e.g., MACD crossover) were deliberately rejected: they would reduce the task to imitating a deterministic rule rather than learning predictive patterns from the feature set.

Table 9: Positive-class rates for buy and sell labels across the six Vietnam tickers.

Ticker	Buy+ (%)	Sell+ (%)
VCB	30.7	26.8
HPG	35.7	30.5
FPT	29.6	25.5
VNM	27.0	21.8
MSN	30.7	28.5
MWG	35.2	25.5

Model Architecture

The classifier reuses the Task 2 LSTM trunk with a sigmoid output head and Dropout regularisation:

Input(30, 12) $\rightarrow$ LSTM(64) $\rightarrow$ Dropout(0.2) $\rightarrow$ Dense(1, sigmoid)

Binary cross-entropy loss was used with the Adam optimiser. A **decision-threshold sweep** was performed on the validation set: the threshold $\theta \in \{0.05, 0.10, \dots, 0.95\}$ that maximised the validation-set F1 score was selected for each ticker independently.

This allows the model to trade off precision and recall according to the asset’s signal characteristics rather than using the default 0.5 cut-off.

Task 3.1 — Buy Signal Results

Table 10: Buy signal classification results on the test set ($h = 5$, $\tau = 0.02$).

Ticker	θ	Acc	Prec	Rec	F1	AUC
VCB	0.25	0.294	0.291	1.000	0.451	0.464
HPG	0.05	0.356	0.356	1.000	0.525	0.463
FPT	0.05	0.314	0.314	1.000	0.478	0.561
VNM	0.05	0.218	0.218	1.000	0.358	0.463
MSN	0.05	0.371	0.371	1.000	0.541	0.503
MWG	0.05	0.339	0.339	1.000	0.506	0.572

Task 3.2 — Sell Signal Results

Buy and sell classifiers are trained as **two independent models** rather than a single three-class softmax. The rationale is twofold: first, the spec explicitly treats them as separate subtasks; second, the two signals are not mutually exclusive in a noisy market — there are windows where neither signal fires (the implicit “hold” regime), which a softmax would force-allocate to one of three classes.

For sell signals, recall is weighted more heavily in the threshold selection: missing a sell signal means holding through a drawdown, which is more costly than a missed buy opportunity in a long-only portfolio.

Table 11: Sell signal classification results on the test set ($h = 5$, $\tau = 0.02$).

Ticker	θ	Acc	Prec	Rec	F1	AUC
VCB	0.05	0.276	0.276	1.000	0.432	0.454
HPG	0.05	0.324	0.324	1.000	0.490	0.616
FPT	0.05	0.217	0.217	1.000	0.357	0.536
VNM	0.05	0.220	0.220	1.000	0.361	0.571
MSN	0.25	0.310	0.308	0.987	0.469	0.504
MWG	0.05	0.373	0.373	1.000	0.544	0.345

Task 4: Portfolio Construction and Risk Management

Task 4 trains no new neural networks. It acts as an **allocation and backtest layer** on top of the models from Tasks 2.3 and 3.2, consuming their test-set outputs to make portfolio decisions.

Task 4.1 — Profitable Stock Selection

Expected-Return Scoring. For each ticker t and each decision day d in the test window, the expected 5-day return is estimated from the Task 2.3 multi-step LSTM:

$$\hat{r}_{t,d} = \frac{\hat{\text{Close}}_{t,d+5}}{\text{Close}_{t,d}} - 1$$

where $\hat{\text{Close}}_{t,d+5}$ is the 5th step of the $k = 7$ multi-step prediction for ticker t (step 5 lies comfortably within the loss-optimised horizon, avoiding extrapolation). The profitability score is then normalised across tickers on each decision day:

$$\text{profit_score}_{t,d} = \frac{\hat{r}_{t,d} - \min_j \hat{r}_{j,d}}{\max_j \hat{r}_{j,d} - \min_j \hat{r}_{j,d} + \epsilon}$$

Table 12 summarises the mean predicted and realised 5-day returns across all test-window decision days, together with the hit rate (fraction of days where the sign of the predicted return matched the realised return) and the cross-day mean normalised rank $\bar{\rho}$.

Table 12: Task 4.1 profitability scoring summary (test window, $h = 5$).

Ticker	N_{test}	Mean $\hat{r}$	Mean real r	Hit rate	Mean rank $\bar{\rho}$
MWG	314	+1.51%	−0.55%	43.9%	−0.127
VCB	502	+0.51%	+0.26%	53.0%	+0.044
FPT	596	+0.32%	+0.79%	56.2%	+0.069
VNM	630	+0.27%	−0.04%	47.9%	−0.044
MSN	489	−0.16%	+0.40%	45.2%	−0.033
HPG	562	−0.21%	+0.30%	45.6%	−0.105

The ranking places MWG first by predicted return, despite a negative realised mean — illustrating that short-horizon LSTM predictions are noisy proxies for actual returns. VCB and FPT rank second and third respectively, and both show positive realised returns, suggesting the model’s signal is partially informative for these tickers.

Task 4.2 — Risk Management

Risk Scoring. The risk score is a z-score blend of two components computed per ticker per day: (1) the 20-day realised volatility σ_{20} , and (2) the sell-signal probability p_{sell} from Task 3.2. Both are standardised then combined, and the composite is min-max scaled across tickers on each decision day so scores lie in $[0, 1]$:

$$\text{risk_score}_{t,d} = \text{MinMax}(z(\sigma_{20,t,d}) + z(p_{\text{sell},t,d}))$$

A high risk score flags a ticker as dangerous to hold. Table 13 summarises the mean risk statistics across the test window.

Table 13: Task 4.2 risk scoring summary (test window). “#Riskiest” = days ranked highest-risk.

Ticker	Mean vol	Mean p_{sell}	Mean risk	#Riskiest	%Riskiest
HPG	0.0243	0.320	0.878	392	62.2%
MSN	0.0247	0.290	0.715	141	22.4%
MWG	0.0238	0.243	0.446	2	0.3%
VCB	0.0176	0.268	0.403	51	8.1%
FPT	0.0172	0.243	0.311	29	4.6%
VNM	0.0145	0.236	0.140	15	2.4%

HPG dominates the riskiest-day count (62% of test days), driven by its high 20-day volatility and sell probability. VNM shows the lowest mean risk score, consistent with its status as a consumer-staples defensive stock.

Investor profiles. Two investor personas are supported:

- **Prudent investor:** risk-aversion weight $\lambda = 2.0$; individual weight cap 25%; the single riskiest ticker (HPG) is hard-excluded.
- **Aggressive investor:** $\lambda = 0.5$; weight cap 50%; no hard-exclusions — all six tickers remain eligible.

Task 4.3 — Portfolio Optimisation and Composition

Allocation strategy. After profitability scoring (Task 4.1) and risk scoring (Task 4.2), a single combined score is computed per (date, ticker):

$$s_{t,d} = \hat{r}_{t,d} - \lambda \cdot \text{risk_score}_{t,d}, \quad s_{t,d} \leftarrow \max(s_{t,d}, 0)$$

Negative scores are clipped to zero so under-water names receive zero weight rather than a short position. Weights are then proportional to the clipped scores and capped at the per-profile maximum. Table 14 summarises the two profile configurations.

Table 14: Portfolio profile parameters.

Profile	λ	Weight cap	Hard-exclude top- K riskiest
Prudent	2.0	25%	$K = 1$ (HPG always excluded)
Aggressive	0.5	50%	None

Both portfolios are benchmarked against an equal-weight allocation over the same six-ticker universe on the same rebalance schedule, so any performance difference is attributable to the scoring rather than the universe choice.

Backtest methodology. The portfolio is evaluated with a walk-forward backtest on the test set, rebalancing every 5 trading days. At each rebalancing point, decisions are made using only information available up to that day (no look-ahead), and positions are held at equal weight until the next rebalancing event.

Portfolio performance metrics:

Table 15: Portfolio backtest results (test window: 2021-11-12 to 2023-02-17, 314 trading days, $r_f = 0$).

Metric	Prudent	Aggressive	Equal-Weight
Annualised Return (%)	−8.33	−9.83	−12.95
Sharpe Ratio	−0.213	−0.288	−0.391
Maximum Drawdown (%)	−24.76	−26.47	−33.51

Discussion

Findings

Multi-feature LSTM vs. single-feature baseline. Extending from a single Open-price feature to the full 13-feature set consistently reduced test-set MSE across all three Nasdaq tickers. The additional technical indicators — particularly MACD and RSI — appear to provide momentum and mean-reversion context that the raw price series alone does not encode.

Error growth with forecast horizon. As expected from financial time-series theory, forecast error grows with k in all tasks. The multi-step model (Task 1.3 / 2.3) exhibits a characteristic concave error growth curve: error increases fastest at early steps (because the model rapidly loses its “anchor” to the last known price) and tapers at later steps (the model converges toward the long-run mean level of the test window).

Classification signal quality. All six buy-signal classifiers converged to near-perfect recall (1.000) at very low decision thresholds ($\theta = 0.05$ for five of six tickers), with VCB being the only exception ($\theta = 0.25$). This indicates the models assign high predicted probability to a large fraction of windows, producing a permissive classifier that captures all positive cases but with precision equal to the positive-class base rate (27–37%). AUC scores above 0.5 for most tickers (up to 0.572 for MWG buy, 0.616 for HPG sell) confirm that the LSTM does learn a discriminative signal, though there is clearly room for improvement.

Sell-signal models show broadly similar patterns, with MSN being the only ticker where the threshold rose to 0.25, suggesting its sell signal is noisier.

Portfolio. All three strategies produced negative annualised returns over the test window (2021-11-12 to 2023-02-17), reflecting the broad drawdown of the Vietnamese market during this period. Nevertheless, the scoring framework delivered meaningful relative outperformance: the Prudent portfolio lost -8.33% annualised (Sharpe -0.213 , max drawdown -24.76%), and the Aggressive portfolio lost -9.83% (Sharpe -0.288 , max drawdown -26.47%), both considerably better than the equal-weight benchmark at -12.95% (Sharpe -0.391 , max drawdown -33.51%). The Prudent profile’s hard-exclusion of HPG — the riskiest ticker by the z-blend score — and its tighter weight cap appear to be the primary driver of its superior drawdown protection (≈ 9 percentage points shallower than the benchmark). The Aggressive profile, while underperforming the Prudent profile, still outperforms the benchmark by over 3 percentage points on both return and Sharpe, confirming that even a loose risk-weighting ($\lambda = 0.5$) adds value relative to naive equal allocation.

Limitations and Future Work

- **Small ticker universe.** Only 3 Nasdaq and 6 Vietnam tickers were modelled. Expanding to sector-level or exchange-wide portfolios would stress-test the pipeline but require substantially more compute.
- **No transaction costs.** The backtest assumes frictionless trading. Real-world bid-ask spreads and brokerage fees would reduce portfolio returns, particularly for the 5-day rebalancing frequency.
- **Hyperparameter sensitivity.** The labelling thresholds $h = 5$, $\tau = 0.02$ and portfolio parameters were set heuristically. A systematic grid search over these values would improve robustness.
- **Alternative architectures.** Transformer-based models (e.g., Temporal Fusion Transformer) could be explored as a comparison point, as their self-attention mechanism may better capture long-range temporal dependencies.

Conclusion

This project demonstrated the end-to-end application of deep learning to stock market time-series data, from raw price history to an actionable portfolio. An LSTM-based architecture with per-window MinMax normalisation and 13 engineered features was shown to forecast Nasdaq and Vietnamese equity prices at multiple horizons. The same architecture, reconfigured as a binary classifier, identified buying and selling signals with

respectable F1 scores on the Vietnam test set. Finally, a two-stage portfolio layer combined the LSTM profitability scores with the sell-risk probabilities to produce both risk-taking and prudent portfolio recommendations.

The project reinforced several practical lessons: chronological data splitting is non-negotiable for financial time-series; per-window normalisation avoids subtle look-ahead leakage; and threshold tuning on a held-out validation set is essential to translate probability outputs into actionable classification decisions.

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